

Distribution Network Topology Error Correction Using Smart Meter Data Analytics

Wenpeng Luan, SMIEEE

wenpeng.luan@bchydro.com

Joshua Peng

joshua.peng@bchydro.com

Mirjana Maras

mirjana.maras@bchydro.com

Joyce Lo

joyce.lo@bchydro.com

Smart Metering Program

BC Hydro

Burnaby, BC, Canada

Abstract - This paper introduces an algorithm which was developed within BC Hydro with the purpose of detecting and correcting topology errors in existing GIS records of the distribution system. Starting with the distribution network topology information available in enterprise GIS, the proposed analytic approach uses smart meter data to detect wrongly connected customers in a transformer neighborhood based on the correlation factors and relative magnitude levels of hourly voltage profiles. Once a wrongly connected customer is identified, further analysis on correlation factors and relative voltage levels is done to correctly reposition this customer within the distribution network topology tree. Testing in a practical BC Hydro distribution system has achieved very encouraging results.

Index Terms—Smart Meters, Advanced Metering Infrastructure (AMI), Data Analytics, Distribution Network Topology.

I. INTRODUCTION

Power companies use the enterprise Geographical Information System (GIS) to keep an inventory of the distribution system assets and how they are connected to each other within the power grid. The network topology defines the connectivity relationships between customers and distribution transformers, as well as distribution transformers and single or multiple phases of distribution feeders. There are many errors in the existing GIS records of the distribution system, including wrong geographical location of assets, mismatching asset parameters, and most importantly incorrect network connectivity. These errors have a significant impact on effective asset management, asset maintenance, outage response, system operations, and workforce safety.

Efforts have been made to reduce the errors in GIS records. Many utilities have developed standard operating procedures to make sure that GIS information is updated when changes in the associated system assets are made. Some utilities organize dedicated line inspection patrols by committing dedicated resources to correct these errors. Power Line Carrier (PLC) technology or GPS (Global Positioning System) clock based phasor measurement technology is sometimes used to detect device connectivity or connectivity relationships with a known system reference. All these initiatives require a lot of time and resources, and thus are not cost effective.

Many power utilities are implementing Advanced Metering Infrastructure (AMI) programs as part of the grid modernization effort. Leveraging two way communication technologies, AMI automatically collects interval smart meter data from end customers to the utility back end [1][2]. Big data resulting from smart meters provides a new means for utilities to improve their GIS records, such as by correlating outage events among meters to determine if they are part of the same feeder section, etc.

This paper introduces an algorithm which was developed within BC Hydro with the aim of using smart meter data to detect and correct distribution system topology errors in the existing GIS records. The algorithm calculates correlation factors and relative magnitude levels of customers' hourly voltage profiles and identifies those customers which were incorrectly placed in the GIS network topology. Further correlation analysis and voltage magnitude profile comparison with neighboring meters in nearby transformers enables the algorithm to reposition those meters into the correct positions inside the network tree. In this way, the algorithm results with a correct distribution network topology tree. Testing to date achieves promising results.

The paper is organized as follows. After background introduction, Section II presents the problem formulation; Section III describes the general approach and rationale behind the proposed algorithm. Detail results from a field test are presented in Section IV and a brief summary is provided in Section V.

II. PROBLEM FORMULATION

Presently, there are many errors in the GIS records of the BC Hydro distribution system. Some of the most common types of errors are:

- Wrong connection points (poles, transformers, feeder phases, feeders, etc.) of devices and wires in the network, the most common examples being customers connected to different distribution transformers, and distribution transformers connected to different feeder phases.

- Incorrect device or wire parameters, such as wire size and length, transformer size, etc.

Errors in the GIS records of the distribution system impact utility planning and operations in many ways. The following lists some of these impacts:

- System analysis and optimization: a wrong topology model will result in wrongly presented load flow in the network, which makes it impossible for system optimization such as Voltage and Var Optimization to achieve the desired performance level.
- Asset management: equipment overload may not be detected and this can result in device failures. Assets can also be under-loaded with a low utilization rate, which reduces the capital investment benefits. Many utilities have experienced this problem with distribution transformers.
- Outage response: wrongly presented network connectivity information may delay the process of restoring power to customers.

III. SMART METER DATA ANALYTIC APPROACH

This section describes the details of the proposed approach for detecting and correcting topology errors in the GIS distribution system records by leveraging and analyzing the big data resulting from smart meter measurements. It can improve the accuracy of network connectivity, which defines the relative position of assets in the network topology tree.

A. Voltage Profile Characteristics in Distribution Systems

Distribution systems are mainly designed to operate in radial topology. Due to load uncertainties and load variations, distribution voltage profiles have the following characteristics:

- Voltage profiles for different feeders are different, since the associated load profiles are different.
- Voltage phase imbalances are common within the same distribution feeder.
- The voltage level (magnitude) decreases in the downstream direction along the feeders.
- Load varies and impacts each other across the system. Loads tend to have similar voltage profile patterns if they are connected to the same network branch or source. The shorter the electrical distance between two loads, the more similar their voltage profiles are. On the other hand, the further away two loads are from each other, the greater the diversity of their voltage profile patterns.

B. Proposed Voltage Profile Analytic Method

A typical urban/suburban distribution system topology, shown in Figure 1, is taken as an example to explain the proposed analytic method. In the system, there are two secondary distribution branches stemming from the distribution transformer, with six customers connected to the vertical branch, and nine customers connected to the horizontal branch.

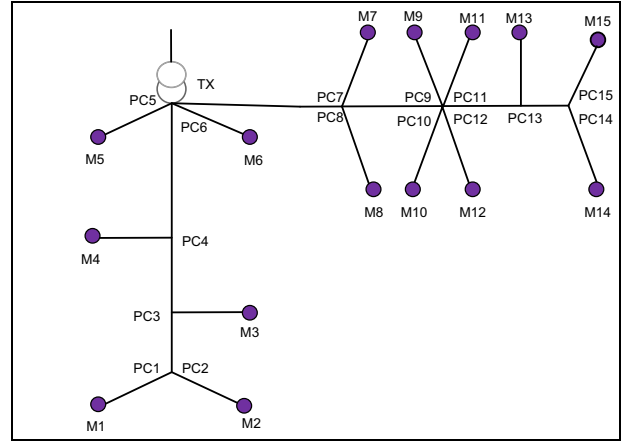


Figure 1. Example urban secondary distribution system

Denoting the connection point of a customer service wire with a secondary distribution branch as the Point of Coupling (PC), one can calculate the voltage at the Point of Coupling, V_{PC} , using customer smart meter measurements and service wire parameters.

According to Ohm's Law, the formula for V_{PC} is:

$$\hat{V}_{PCj} = V_j + Z_{sj} \cdot \hat{I}_j, \quad (1)$$

where j is the meter number associated with a particular customer, $\hat{I}_j$ is the hourly current recorded by the smart meter j , V_j is the average hourly voltage recorded by the smart meter j , and Z_{sj} is the impedance of the service wire for this particular meter.

For smart meters which do not have $\hat{I}_j$ available, but do have hourly average voltage V_j , and hourly energy consumption kWh, the hourly meter load current $\hat{I}_j$ can be approximated by kWh/V_j .

The voltage profile recorded by a customer's smart meter accounts for the voltage drop along the service wire which often vary in length for different customers. Calculation of V_{PCj} allows for a meaningful comparison of voltage profiles evaluated at points of coupling for all meters connected to the same secondary branch. Due to the influence of the source voltage profile, these computed voltage profiles should have similar patterns over time, and differ in their magnitudes. The voltage magnitude of downstream meters should be lower than that of upstream meters, due to the technical losses along the branch. It is obvious that voltage is highest at the distribution transformer and lowest at the branch end, and voltages are same for neighbours share the same point of coupling.

A simplified view of the demonstration system described in Figure 1 is shown in Figure 2.

Considering the vertical branch shown in Figure 2, the magnitude of V_{PCj} values over time has the following ranking from the highest to the lowest magnitudes:

$$V_{PC5} = V_{PC6} > V_{PC4} > V_{PC3} > V_{PC1} = V_{PC2}. \quad (2)$$

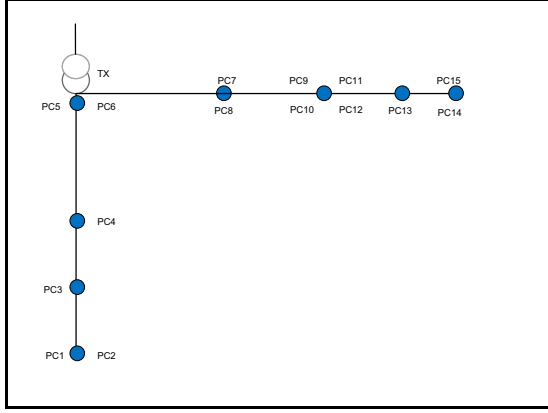


Figure 2. Simplified network topology tree

Similarly, the V_{PCj} values along the horizontal branch should follow the following ascending order:

$$V_{PC5} = V_{PC6} > V_{PC7} = V_{PC8} > V_{PC9} = V_{PC10} = V_{PC11} = V_{PC12} > V_{PC13} > V_{PC14} = V_{PC15}. \quad (3)$$

The proposed algorithm performs the following computations and analysis:

1. Calculate the voltage profile V_{PCj} at the point of coupling PCj for each meter j , as described in (1) based on the following data: service wire impedance, hourly voltage and energy measurements for a continuous period (e.g., 1-2 weeks);
2. Carry out statistical correlation analysis of the V_{PCj} voltage series of all meters in a defined geographical area, which includes all meters on the same transformer and neighboring transformers within identified proximity. The result is an $N \times N$ correlation matrix which indicates the correlation factors, $\text{corr}(V_{PCj}, V_{PCk})$, for $j, k = \{1, 2, \dots, N\}$, where N is the total number of meters considered in the analysis.
3. Use the computed correlation factors to assess the electrical distance between each pair of meters. For instance, the meters with the highest correlation are expected to be direct neighbors, either by sharing the same point of coupling, or by being directly downstream or upstream from each other. The greater the electrical distance, the smaller the correlation factor is expected to be. Furthermore, V_{PCj} magnitude will indicate the meter's downstream/upstream position in comparison to other nodes in the distribution network topology tree, as shown in Figure 2. If both the correlation and magnitude of voltage profiles conform to the network topology recorded in GIS, then the network topology is expected to be correct.
4. If an anomaly is found with an existing connection relationship recorded in GIS, for instance, if V_{PCj} has relatively low correlation with its neighboring

meters on the same transformer, this may indicate that meter j is connected to a different transformer. The strongest correlation among all the evaluated meters should identify its actual neighbors with least electrical distance in the system. When the V_{PCj} profile pattern identifies the specific meter group the meter is in, its voltage magnitude can be used to position the meter into the correct location on the network branch, as shown in Figure 2.

With this approach, the correct network topology can be reconstructed after a topology error has been detected.

IV. TESTING RESULTS

The proposed method is tested in an urban area of BC Hydro's distribution system. It includes 84 residential customers supplied by 8 single phase distribution transformers, which are connected to 2 feeders (phase C of feeder 1, and phase A, B, or C of feeder 2). All topology errors were correctly detected by the algorithm, which were further confirmed by a site visit. The results are presented in this section. For simplicity, some of the meters under listed transformers are omitted in the presentation of results.

A. Scenario 1: Individual Meter on a Wrong Transformer

The first scenario pertains to transformer 1 on Feeder 1, phase C, which according to GIS records has 13 customers connected to it. The resulting correlation factors calculated with one week's voltage profile of all meters in this transformer are presented in Table I, which shows that all meters other than 11 and 13 are correlated with each other with correlation factors greater than 0.97. It indicates that these meters are connected to same transformer 1, while meters 11 and 13, which correlate with other meters in the low 0.67-0.78 range, are highlighted to be anomalous meters, which may actually be connected to different transformer.

TABLE I: CORRELATION RESULTS FOR TX1

Meter ID	m1	m2	m3	m4	m5	m6	m7	m8	m9	m10	m11	m12	m13
m1	1	0.990	0.994	0.993	0.995	0.995	0.990	0.990	0.986	0.989	0.690	0.994	0.766
m2	0.990	1.000	0.992	0.996	0.993	0.993	0.983	0.985	0.977	0.982	0.714	0.988	0.782
m3	0.994	0.992	1.000	0.996	0.999	0.997	0.986	0.987	0.980	0.986	0.707	0.993	0.768
m4	0.993	0.996	0.996	1.000	0.997	0.997	0.987	0.988	0.981	0.984	0.710	0.992	0.776
m5	0.995	0.993	0.999	0.997	1.000	0.999	0.991	0.992	0.986	0.990	0.710	0.996	0.775
m6	0.995	0.993	0.997	0.997	0.999	1.000	0.992	0.993	0.987	0.990	0.707	0.997	0.774
m7	0.990	0.983	0.986	0.987	0.991	0.992	1.000	0.997	0.994	0.996	0.685	0.998	0.768
m8	0.990	0.985	0.987	0.988	0.992	0.993	0.997	1.000	0.993	0.995	0.691	0.997	0.772
m9	0.986	0.977	0.980	0.981	0.986	0.987	0.994	0.993	1.000	0.992	0.681	0.994	0.762
m10	0.989	0.982	0.986	0.984	0.990	0.990	0.996	0.995	0.992	1.000	0.673	0.997	0.763
m11	0.690	0.714	0.707	0.710	0.710	0.707	0.685	0.691	0.681	0.673	1.000	0.693	0.932
m12	0.994	0.988	0.993	0.992	0.996	0.997	0.998	0.997	0.994	0.997	0.693	1.000	0.769
m13	0.766	0.782	0.768	0.776	0.775	0.774	0.768	0.772	0.762	0.763	0.932	0.769	1.000

Figure 3 shows the voltage profiles for meters on transformer 1 for a single representative day. One can easily see that meters 11 and 13 have much higher voltage levels and different shapes from the voltage profiles of other meters. This agrees with the prediction that these two meters are connected to a different transformer.

In order to determine which transformers these two meters are connected to, a correlation study was performed on the

two meters and their neighboring meters which are connected to the geographically neighboring transformers (TX2 and TX3). The results are shown in Table II.

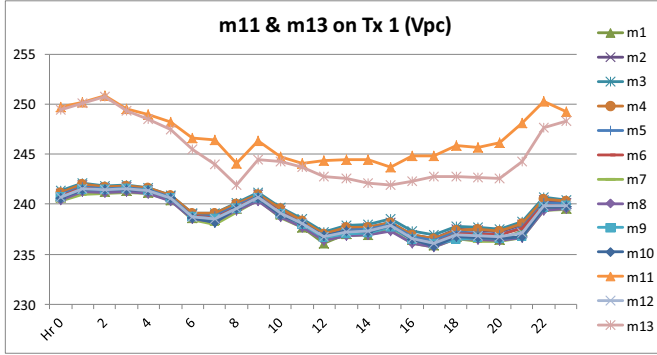


Figure 3. Transformer 1 voltage profiles

TABLE II: CORRELATION RESULTS FOR TX2 & TX3

Meter ID	m11	m13	m14	m15	m16	m17	m18	m19	m20	m21	m22
m11	1.000	0.932	0.997	0.993	0.994	0.990	0.932	0.935	0.944	0.950	0.949
m13	0.932	1.000	0.935	0.934	0.933	0.931	0.997	0.997	0.995	0.989	0.989
m14	0.997	0.935	1.000	0.998	0.999	0.996	0.935	0.938	0.947	0.953	0.952
m15	0.993	0.934	0.998	1.000	0.998	0.994	0.933	0.937	0.945	0.951	0.949
m16	0.994	0.933	0.999	0.998	1.000	0.995	0.933	0.936	0.945	0.951	0.949
m17	0.990	0.931	0.996	0.994	0.995	1.000	0.931	0.934	0.944	0.952	0.950
m18	0.932	0.997	0.935	0.933	0.933	0.931	1.000	0.996	0.996	0.990	0.990
m19	0.935	0.997	0.938	0.937	0.936	0.934	0.996	1.000	0.996	0.989	0.990
m20	0.944	0.995	0.947	0.945	0.945	0.944	0.996	0.996	1.000	0.995	0.997
m21	0.950	0.989	0.953	0.951	0.951	0.952	0.990	0.989	0.995	1.000	0.996
m22	0.949	0.989	0.952	0.949	0.949	0.950	0.990	0.990	0.997	0.996	1.000

Results presented in Table II indicate that the meter 11 is correlated most strongly with the meter m14 on transformer 2, and that the meter 13 is correlated most strongly with meters 18 and 19 on transformer 3. This identifies their closest neighbors (with least electrical distance), and thus indicates their actual connections.

Given the results of the correlation analysis, the daily V_{PC} profiles of the meter m11 and the meter m13 are compared with the corresponding daily V_{PC} profiles of the meters in the neighboring transformer 2 and transformer 3, respectively. These comparisons are shown in Figures 4 and 5.

Figure 4 clearly shows that meter m11 belongs to transformer 2, as its V_{PC} profile follows very closely the V_{PC} variations of other meters in that group. The V_{PC} magnitude of the meter m11 indicates that it is connected at the end of the secondary branch, just next to meter 14. For comparison purpose, the voltage profile of meter 13 is also shown in this graph, and it clearly indicates that meter 13 is not connected to transformer 2.

Similarly, Figure 5 shows that meter 13 has the same V_{PC} pattern as the meters on transformer 3. With the lowest V_{PC} value, meter 13 is predicted to be connected in the most downstream position on transformer 3, downstream of meters 18 and 19.

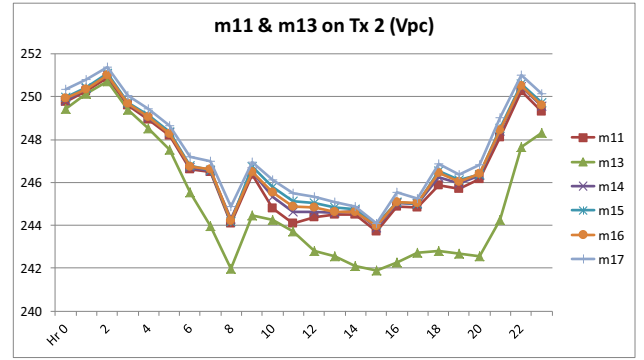


Figure 4. Voltage profile comparison with Transformer 2

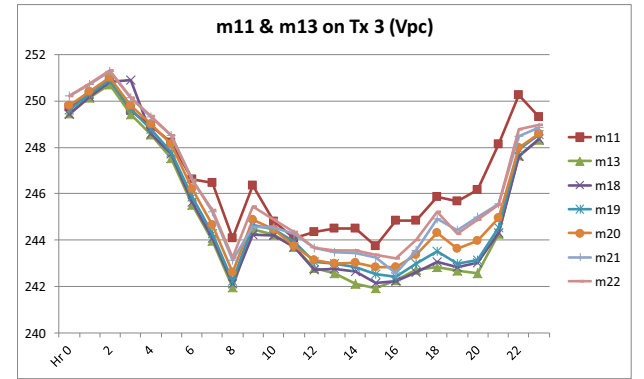


Figure 5. Voltage profile comparison with Transformer 3

Figure 6 shows the original GIS connectivity represented in blue lines, with the topology error corrections as verified in the field in red lines. In the case of meter 11, it is connected directly downstream of meter 14 on transformer 2, but was wrongly recorded to transformer 1. In the case of meter 13, it is placed directly downstream of meters 18 and 19 of transformer 3. Both cases match former predictions.

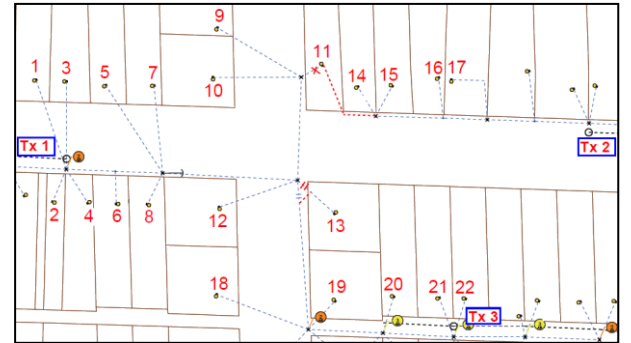


Figure 6. Actual network connections

B. Scenario 2: A Group of Meters on a Wrong Transformer

A correlation study was also performed on two neighboring transformers, namely transformer 4 and transformer 5, which both belong to the Feeder 2, phase C, with transformer 5 being upstream from transformer 4. Table 3 contains the correlation factors between V_{PC} values of the meters on these two transformers and shows that all correlation factors have high values. The four meters 29-32 on transformer 5 correlate well with each other, so there is no

doubt that they belong to the same transformer. However, their correlation factors with meter 28, which according to GIS records belongs to transformer 4, indicate that meter 28 is their direct neighbor, followed by meters 26 and 27 as their subsequent neighbors, and meters 23-25 as the furthest neighbors. These results indicate that the meters belonging to transformer 5 should be on transformer 4, and positioned at the end of the secondary branch on this transformer.

TABLE III CORRELATION RESULTS WITH TX 4 & TX 5

Meter ID	m23	m24	m25	m26	m27	m28	m29	m30	m31	m32
m23	1.000	0.999	0.999	0.992	0.985	0.981	0.966	0.967	0.964	0.973
m24	0.999	1.000	0.998	0.991	0.984	0.981	0.966	0.968	0.965	0.974
m25	0.999	0.998	1.000	0.992	0.986	0.983	0.968	0.969	0.966	0.975
m26	0.992	0.991	0.992	1.000	0.998	0.996	0.987	0.988	0.986	0.992
m27	0.985	0.984	0.986	0.998	1.000	0.995	0.986	0.988	0.986	0.991
m28	0.981	0.981	0.983	0.996	0.995	1.000	0.995	0.996	0.995	0.998
m29	0.966	0.966	0.968	0.987	0.986	0.995	1.000	1.000	1.000	0.999
m30	0.967	0.968	0.969	0.988	0.988	0.996	1.000	1.000	1.000	0.999
m31	0.964	0.965	0.966	0.986	0.986	0.995	1.000	1.000	1.000	0.999
m32	0.973	0.974	0.975	0.992	0.991	0.998	0.999	0.999	0.999	1.000

The V_{PC} profiles further confirm this prediction. Figure 7 shows that V_{PC} levels of the meter group 29-32 are lower than those of the meter group 23-28.

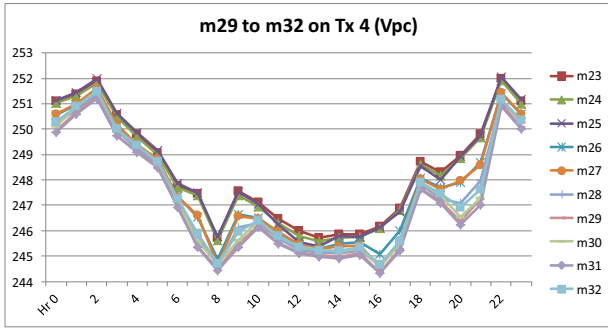


Figure 7. Voltage profile comparison with Transformer 4

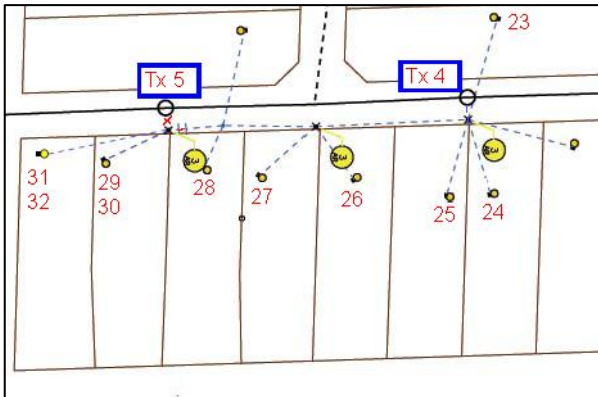


Figure 8. Correct connections associated with transformers 4 and 5

Site visit found that transformer 5 supplies no customer other than a communication device, and transformer 4 supplies all meters 23-32. The correct connectivity diagram is shown in Figure 8.

C. Scenario 3: Transformer on a Wrong Feeder Phase

The study area also contained two transformers on Feeder 2, connected to phase A and phase B separately. However, a correlation study showed that their voltage profiles correlate with an average correlation factor of 0.98, which is unusual for different feeder phases. A site visit confirmed that both transformers belong to the Feeder 2, phase B.

D. Discussion

Further investigation found that the transformer 4 described in Scenario 2 was identified as having a “90.26%” load factor in the asset inventory system. With the additional four meters being identified, this transformer would be overloaded during peak time. This showcases the value of topology error correction with regards to asset management.

Scenario 3 identified two transformers to be on the same feeder phase, however it was difficult to determine if they were connected to phase A or B without a site visit. BC Hydro is planning to deploy feeder meters as a part of SMI implementation, and these can be used as a reference for feeder phase identification, thus solving this problem.

Correlation analysis was also performed with the hourly average voltage measurements (V_h) directly acquired from smart meters, which resulted in slightly degraded but acceptable correlation factors. However, the V_{PC} approach enables more reliable and efficient rebuilding of the correct topology tree.

The results presented in this paper pertain to the secondary system; however the same algorithm can be easily extended to the primary distribution system. Further analysis is being performed at this time.

V. SUMMARY

This paper introduced an algorithm developed within BC Hydro with the purpose of using smart meter data to detect and correct distribution system topology errors in existing GIS records. By evaluating the correlation factors and relative magnitude levels of customers’ hourly voltage profiles at points of coupling along the secondary branch of distribution transformers, the algorithm identifies those customers which were incorrectly placed in the GIS network topology and uses further analysis to. In this way, the algorithm reconstructs the correct distribution network topology tree by repositioning these wrongly connected customers into the correct positions. Testing results prove its efficiency. It is expected that the same algorithm can be applied to the primary distribution system.

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